

Regenerative Braking

by

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Abstract

This project investigates regenerative braking for small electric mobility platforms (e.g., scooters and e-bikes) through the design of a bench-scale demonstrator. The proposed system uses a three-phase brushless DC (BLDC) hub motor, a custom three-phase inverter, and a buck converter that transfers recovered energy into a supercapacitor bank during braking. To minimize cost and reduce safety risk, the prototype operates from a 7.5 V, 6 A supply (≈ 45 W maximum). A nominal regenerative power target of 25 W was selected to produce clearly observable deceleration and a measurable rise in supercapacitor voltage while remaining within supply constraints.

Analytical sizing and MATLAB simulations were used to predict braking behavior and capacitor charging dynamics. The hardware architecture was implemented around an ESP32 microcontroller generating PWM for both inverter commutation and buck charging control, with Hall sensor feedback and shunt-based current measurement for protection. During final integration, the team validated key control signals and developed a staged test plan; however, limitations of breadboard-level power electronics prototyping (contact resistance, current handling, and device packaging constraints) prevented reliable motoring of the hub motor within the available time.

The report documents the final architecture, calculations, prototype implementation, and a traceable validation strategy, and concludes with recommended next steps to achieve full motoring and regenerative braking validation using a PCB or power prototyping platform.

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Nomenclature

Symbol	Definition	Units
V_{bus}	DC bus voltage	V
I_{bus}	DC bus current	A
P_{regen}	Regenerative power transferred to storage	W
η	Converter efficiency	–
K_e	Back-EMF constant (approx.)	V/rpm
C_{eq}	Equivalent supercapacitor capacitance (series bank)	F
E_{cap}	Energy stored in capacitor bank	J
D	Buck duty cycle	–
f_s	Switching frequency	Hz
ΔI_L	Inductor ripple current (peak-to-peak)	A
L	Buck inductor value	H

Nomenclature and symbol definitions used throughout the report.

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1 Introduction/Background

1.1 Problem Statement

Mechanical friction brakes dissipate a vehicle's kinetic energy as heat. In small electric mobility platforms such as scooters and e-bikes, this means that a potentially recoverable energy stream is usually lost, even though the traction motor can operate as a generator during deceleration. Regenerative braking is well established in larger electric vehicles, but small low-power platforms still face integration barriers related to cost, controller capability, battery charging limits, and the lack of accessible test platforms for repeatable laboratory validation.

The engineering gap addressed in this project is therefore not simply whether regenerative braking is theoretically possible, but whether a low voltage, low power, single motor system can be implemented in a safe and measurable way for bench-scale testing. The project focuses on a 24 V class system that is representative of scooter-scale hardware while remaining suitable for educational use, instrumentation, and incremental design refinement.

The specific problem statement for the remainder of this report is: design and validate a bench-scale regenerative braking prototype using a 7.5 V 6A battery using one direct-drive BLDC motor for both propulsion and braking, while demonstrating controlled negative torque, safe battery-side energy return, and measurable recovered energy using a capacitor under repeatable test conditions.

1.2 Design Requirements

ID	Requirement	Target / constraint	Verification method
R1	Battery Selection	Maximum operating current of 6A	Measure with Multimeter
R2	Motor selection	Single direct-drive BLDC hub motor, 200W class, Hall-sensored	Datasheet and bench integration
R3	Regenerative current command	Adjustable battery charging current in the initial range of 0.5-3 A	Current sensor logging
R4	Braking controllability	Smooth commanded deceleration with ramp-limited regen request	Speed trace and command response
R5	Target Regenerative efficiency	$\geq 10\%$	Measure the increase in supercapacitor energy during a braking event and compare it to the estimated drop in wheel/motor kinetic energy from the speed change.
R6	Protection limits	Respect BMS charge limit and controller current limit; include over-voltage shutdown	Fault tests at conservative limits
R7	Data acquisition	Log voltage, current, and wheel speed at ≥ 50 Hz with synchronized timestamps	CSV log review
R8	Mechanical safety	Rigid stand, wheel guard, and hardware emergency stop	Inspection and safety checklist
R9	Prototype form factor	Bench-scale, modular assembly that can be instrumented and disassembled	Assembly review
R10	Budget	Prototype target $\leq \$350$	Budget table and receipts
R11	Maximum supercapacitor voltage	$\leq 2.7V$	Measure supercapacitor voltage during braking/charging with a multimeter/ADC. Confirm it never exceeds 2.7 V and that charging disables at the limit.
R12	Energy recovery per braking event	2-5 J	Record Capacitor voltage before & after braking event & calculate recovered energy from voltage rise

R13	Control Signal Voltage	5 V logic	Use an oscilloscope to verify the PWM logic-high level into the gate-driver inputs is ~5 V
R14	Switching Capability	PWM based inverter control	Use an oscilloscope to confirm PWM is present on correct phase control lines and that the 6 step commutation sequence changes correctly with Hall sensor state
R15	Response Time (Braking Activation)	< 100 ms	Use an oscilloscope to measure time from brake command to inverter PWM shutoff
R16	Gate Drive Voltage	10V minimum to supply MOSFETS	Probe gate-to-source voltage with an oscilloscope
R17	Voltage Regulation for Capacitor	Buck Converter to step down voltage to prevent capacitor voltage exceeding 2.7V	Use an oscilloscope/multimeter to monitor Vcap while enabling the buck

1.3 Background Information/ Detailed Literature Review

Recent literature consistently identifies regenerative braking as one of the most effective ways to improve electric-drive efficiency, particularly in stop-and-go duty cycles [1]-[4], [7], [9]. For light electric vehicles, drivetrain topology strongly affects practical recoverability. Direct-drive hub motors preserve bidirectional torque flow more naturally than geared systems because they avoid one-way freewheel mechanisms and additional transmission losses [5], [6]. Brushless DC and permanent-magnet motor platforms dominate current micromobility applications because of their high efficiency, favorable torque density, and low maintenance burden [3], [5], [6].

However, much of the recent regenerative braking literature is oriented toward vehicle-level control allocation, multi-motor coordination, or high-performance energy-storage optimization rather than low-cost, openly instrumented educational prototypes [7]-[10].

Experimental papers also tend to assume access to specialized vehicle platforms or advanced power electronics that are not directly transferable to a student bench demonstrator [4], [8], [10]. The reviewed work suggests three consistent design lessons. First, energy recovery is only useful if the controller can regulate braking torque and prevent DC-bus over-voltage [2], [4], [7], [9]. Second, battery acceptance and BMS behavior constrain how aggressively energy can be returned to storage, especially near full state of charge [7], [9], [10].

Third, a direct-drive BLDC architecture remains the most representative single-motor topology for scooter-scale implementations [5], [6]. The gap that motivates this project is therefore a practical one: there is still value in developing a modular, bench-scale regenerative braking platform that demonstrates the full energy path from wheel deceleration to battery charging using affordable parts, measurable safety limits, and repeatable instrumentation. This report is revised around that gap rather than around a broad descriptive overview of regenerative braking technology.

1.4 Project Objectives

1. Develop a three-phase inverter using discrete MOSFETs and gate driver ICs capable of driving a 200 W BLDC motor.
2. Configure the system such that kinetic energy from the motor during deceleration can be converted into electrical energy through controlled switching of the inverter.
3. Integrate a 10F, 2.7 V supercapacitor with appropriate voltage regulation to safely capture regenerated energy.
4. Use an Arduino-based platform to generate PWM control signals for the inverter and coordinate system operation.
5. Verify correct operation of gate drivers, switching signals, and circuit behavior using measurement tools such as an oscilloscope.
6. Evaluate the system based on expected energy recovery, electrical behavior, and observed hardware constraints, and identify key factors limiting full system operation.
7. Ensure that the total system cost remains within the defined budget while maintaining functional feasibility and scalability.

2 Design Approach

2.1 Concept Generation

Several system-level approaches were considered before finalizing the prototype architecture. The purpose of this section is to show how each concept addresses the core project goals: measurable energy recovery, safe control, affordability, and relevance to modern light electric vehicles.

Concept A - BLDC motor with a custom three phase controller. This concept uses a direct-drive BLDC hub motor and a controller capable of both propulsion and regenerative braking. It best matches modern scooter and e-bike electrical architectures and preserves the desired single-motor energy path, as well as providing a cheap alternative to a speed controller in comparison to buying a commercial one. Its main challenge is implementation complexity and high technical depth.

Concept B - Brushed hub motor with four-quadrant H-bridge. This concept simplifies motor control and lowers entry cost, but it is less representative of contemporary micromobility powertrains and introduces a maintenance penalty associated with brushed machines. It remains a useful comparison concept because it highlights how lower cost can be traded against architectural relevance and long-term efficiency.

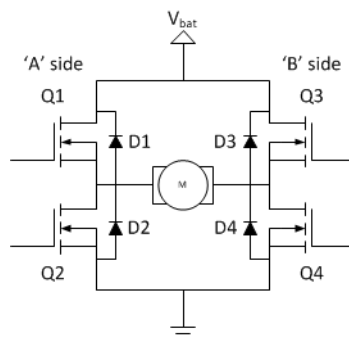


Figure 1. Illustrative four-quadrant H-bridge topology considered for Concept B.

Concept C - Rectifier plus DC-DC converter. In this concept, the back-driven BLDC motor feeds a rectifier and a regulated DC-DC stage that controls the energy returned to the storage bus. This approach provides flexibility in current regulation and voltage matching but does not, by itself, solve the complete propulsion problem. It is therefore more suitable as an energy-recovery subsystem than as a complete drivetrain architecture.

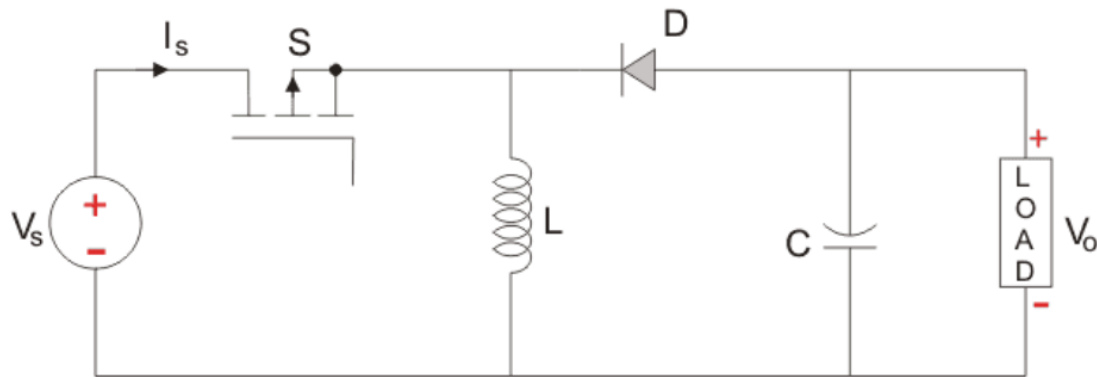


Figure 2. Illustrative buck-boost energy-conditioning stage considered for Concept C.

Concept D - Separate motor and generator. This option was rejected early because it adds cost and departs from realistic single-machine electric vehicle layouts. It would demonstrate energy conversion, but not the desired dual-use operation of one traction machine.

Concept E - Direct implementation on an e-bike. Although realistic, this option introduces uncontrolled environmental variables, more demanding mechanical integration, and a higher safety burden. The team therefore retained a bench-scale demonstrator as the main path for repeatable midterm and final validation.

Concept	Main architecture	Primary strength	Primary limitation
A	Direct-drive BLDC + three-phase controller	Most representative of target platform	Higher integration complexity
B	Brushed hub + four-quadrant H-bridge	Lower controller cost and simpler control	Lower relevance to modern platforms
C	BLDC rectifier + DC-DC stage	Good current-conditioning flexibility	Not a complete propulsion solution
D	Separate motor and generator	Simple demonstration of energy flow	Extra machine cost and poor realism
E	Direct e-bike implementation	Most realistic operating environment	Low repeatability and higher safety burden

2.2 Concept Evaluation and Final Concept Selection

Criterion	Weight	Concept A	Concept B	Concept C
Implementation Complexity	0.20	5	2	3
Demonstrability of regen	0.15	4	4	3
Safety / bounded operation	0.15	3	3	2
Braking controllability	0.15	4	3	2
Feasibility within schedule	0.15	3	4	2
Prototype cost	0.10	2	4	3
Engineering learning / originality	0.10	5	3	4
Weighted total	1.00	3.85	3.10	2.75

With the revised weighting, Concept A becomes the highest-ranked option because it best satisfies the project's central design intent: a single-motor, production-relevant regenerative braking architecture. The selected implementation path is intentionally staged. Phase 1 uses an off-the-shelf regenerative BLDC controller to validate braking torque response, battery acceptance, and data logging. Phase 2 extends the platform toward a custom inverter and control implementation. Although simpler concepts offered advantages in implementation, Concept A was selected due to higher potential efficiency, greater engineering depth, and scalability.

3 Design Analysis

3.1 Engineering Techniques/Software tools

This explicit tool selection answers how the prototype will be engineered:

Tool / platform	Role in project
MATLAB / Simulink	First-order motor, current, and regenerative braking modelling; comparison against measured traces
ESP32 programmed in C++ (Arduino framework)	Brake-input processing, supervisory control logic, sensor acquisition, CSV data logging
Python	Post-processing of logged data, plotting, numerical integration of recovered energy, and report graphics
SolidWorks	Bench-stand layout, guard dimensions, and mounting checks for safe mechanical assembly
Altium	Custom inverter and sensing-board schematic / PCB development if the project advances beyond the commercial controller

As shown, the ESP32 provides the supervisory layer and data logger, MATLAB/Simulink supports prediction and comparison, CAD supports the physical rig, and Altium is reserved for the higher-risk custom power-electronics extension.

3.2 Final Analysis/Calculations

Overview and constraints

The prototype demonstrates regenerative braking by disabling motoring PWM and enabling a buck charger that draws power from the DC bus and stores it in a supercapacitor bank. The power source is a 7.5 V, 6 A adapter, so all target power levels are chosen to remain below this limit with margin for losses. It should be noted that the BLDC motor used in this project is rated for 24 V operation; however, the available supply voltage is only 7.5 V. This significant voltage mismatch reduces the achievable motor torque and speed, and consequently limits the generated back electromotive force (EMF), which is critical for regenerative braking.

$$P_{\max} = V_{\text{bus}} \cdot I_{\max}$$

With $V_{\text{bus}} = 7.5 \text{ V}$ and $I_{\max} = 6 \text{ A}$, $P_{\max} \approx 45 \text{ W}$.

1) Regen/braking power target (chosen)

Nominal regen/braking target of 25 W (adjustable 10–40 W). This is high enough for visible braking and a measurable capacitor voltage rise, while staying well below the 45 W ceiling.

$$P_{\text{regen}} \approx V_{\text{bus}} \cdot I_{\text{bus}}$$

Assuming buck efficiency $\eta \approx 0.9$, the bus current needed is:

$$I_{\text{bus,target}} \approx P_{\text{regen,target}} / (\eta \cdot V_{\text{bus}})$$

For $P_{\text{regen,target}} = 25 \text{ W}$, $I_{\text{bus,target}} = 25 / (0.9 \cdot 7.5) = 3.7 \text{ A}$ (within the adapter limit).

This becomes the primary “brake strength” control variable: higher commanded buck current leads to higher bus current draw, higher generator current, and stronger opposing torque.

2) Regen feasibility (back-EMF argument)

Back-EMF is approximately proportional to speed. Using the rated point (about 24 V at 900 rpm), a rough estimate is:

$$K_e \approx 24 \text{ V} / 900 \text{ rpm} \approx 0.0267 \text{ V/rpm}$$

Regen is strongest at higher RPM; as the motor slows, back-EMF drops and regen current naturally decreases.

To push energy onto a 7.5 V DC bus through inverter body diodes (and real conduction drops), regen becomes significant once the motor-generated line voltage is on the order of ~8 V equivalent. Given that the available supply voltage is substantially lower than the motor’s rated voltage, the system may not reach the rotational speeds required to generate sufficient back-EMF

for effective regenerative braking. This further constrains the ability of the system to operate within the intended regenerative region.

This implies an approximate regen onset speed:

$$\text{rpm_regen} \approx 8 / 0.0267 \approx 300 \text{ rpm.}$$

Therefore, regen energy capture is expected to be effective at moderate-to-high RPM, and will taper off at low speed. This behavior is acceptable for a demonstration because it naturally produces a deceleration curve: strong braking initially, weaker braking near stop.

3) Supercapacitor bank (chosen) and expected voltage rise

To keep currents reasonable and produce a clearly measurable voltage rise, we choose a 5.4 V supercapacitor bank: two 2.7 V EDLC cells in series. Selecting 10 F cell gives an equivalent capacitance $C_{eq} = 10 \text{ F}$. Balancing resistors (e.g., $100 \text{ k}\Omega$ across each cell) are included.

Capacitor energy:

$$E_{cap} = 1/2 \cdot C_{eq} \cdot V_{cap}^2$$

Predicted capacitor voltage after a braking interval t :

$$V_2 = \sqrt{V_1^2 + 2 \cdot P_{\text{regen,target}} \cdot t / C_{eq}}$$

With $C_{eq} = 10 \text{ F}$ and a 5 s braking interval at 25 W starting from $V_1 = 0 \text{ V}$, the model predicts $V_2 \approx 5 \text{ V}$, which is easily measurable during a lab demo. This prediction assumes that sufficient rotational energy is available from the motor, which is dependent on achieving adequate operating speed.

4) Buck converter sizing (chosen)

Choosing a 20 kHz switching frequency (above audible range and compatible with timer-based PWM). Duty cycle varies with capacitor voltage:

$$D \approx V_{\text{cap}} / V_{\text{bus}}$$

For safe ramp-up testing, limit buck currents to $I_{\text{in,limit}} = 4 \text{ A}$ and $I_{\text{out,limit}} = 6 \text{ A}$. Inductor ripple is chosen as $\Delta I_L \approx 30\%$ of $I_{\text{out,limit}}$.

$$\Delta I_L = 0.30 \cdot I_{\text{out,limit}} = 0.30 \cdot 6 \approx 1.8 \text{ A}.$$

Inductor sizing is:

$$L = ((V_{\text{in}} - V_{\text{out}}) \cdot D) / (\Delta I_L \cdot f_s)$$

Using $V_{\text{in}} = 7.5 \text{ V}$ and representative $V_{\text{out}} \approx 5.0 \text{ V}$, $D = V_{\text{out}}/V_{\text{in}} = 5.0/7.5 = 0.667$. This gives $L \approx 46 \mu\text{H}$, so choose $L = 47 \mu\text{H}$ with $\geq 10 \text{ A}$ saturation current margin since:

$$I_{L,\text{peak}} = I_{\text{out,limit}} + \Delta I_L/2 = 6 + 1.8/2 = 6.9 \text{ A}$$

Local capacitors (buck + gate driver):

A $0.1 \mu\text{F}$ ceramic decoupling capacitor is placed directly across the gate driver supply ($V_{\text{CC}} - \text{COM}$) to suppress switching noise, and a $0.1 \mu\text{F}$ ceramic bootstrap capacitor is used between $V_{\text{B}} - V_{\text{S}}$ to support the high-side driver. For the buck power stage, a bulk output capacitor of approximately $980 \mu\text{F}$ is used to reduce output ripple and stabilize the converter during supercapacitor charging.

5) Loss/limit checks (short feasibility checks)

Blocking diode loss (adapter isolation during regen):

$$P_{\text{diode}} \approx I_{\text{bus}} \cdot V_{\text{f}}$$

Current sensing uses shunt resistors ($V_{\text{shunt}} = I \cdot R_{\text{shunt}}$) to detect overcurrent and enter a fault state (PWM disabled). ESR heating in the supercap bank is checked using $P_{\text{ESR}} = I_{\text{cap}}^2 \cdot R_{\text{ESR}}$ to set safe current limits.

Simulations

The primary focus of preliminary testing for the regenerative braking system has been to validate the theoretical behavior of the system through simulation. All simulations were conducted in MATLAB, which allowed us to better understand the expected dynamics of the system prior to building and testing a complete hardware prototype. Each of these simulations' models how the motor's rotational dynamics interact with the electrical charging characteristics of the supercapacitor.

The simulation included three main variables: angular speed of the motor over time, regenerative current generated during braking, and voltage increase across the supercapacitor during energy recovery. The simulation assumes a hub motor with an approximate rated speed of 900 RPM, and the hub motor has an approximate maximum experiment operational speed of 1500 RPM. The estimated moment of inertia for the motor and wheel assembly is approximately $0.006 \text{ kg} \cdot \text{m}^2$, based on the measured wheel radius and motor weight.

According to the simulation results, when the motor can start regenerative braking, it will begin at about 1,500 RPM after which the angular speed will gradually decrease until equilibrium speed is reached because the electromagnetic braking torque produced by the generator action of the motor acts on its own angular momentum. The speed at which the motor operates when it generates the highest braking torque is when it is operating at the highest possible speed: this brake torque is directly proportional to the voltage produced by the motor,

which is also directly proportional to the speed of the motor's rotation and, in general, for the dc motor to produce the highest voltage, it needs to be turning quickly. When the speed of the motor decreases, the voltage produced by the motor also decreases, and hence, the regenerative current produced by the motor also decreases. Therefore, the result of this relationship is that as the motor slows down, it creates an appropriate deceleration curve that will initially be fairly steep, and as the motor approaches lower speeds, the deceleration curve will become increasingly flat.

The next output of the simulation tracked the regenerative charging current into the large supercapacitor. During the initial braking operation of the motor, the charging current to the large supercapacitor will be at or near its peak charging rate (approximately 1.5 A to 2 A depending on how you have configured the input settings) at the time the motor's rotor rotates at maximum speed. Once the rotation of the rotor begins to slow down, the charging current into the supercapacitor will also slow down due to the increasing voltage on the supercapacitor. When the voltage on the supercapacitor nears the voltage generated by the motor, the regenerative charging current will approach zero and the transfer of regenerative energy will cease.

This is how a generator charges a storage device. The final simulated test gave us more information about the amount of voltage rise across a supercapacitor during a regenerative braking process. The supercapacitor reached a maximum value of about 50 F during the simulation. When we started from zero volts on the supercapacitor, the simulation showed that the voltage was continuously increasing throughout the whole duration of the braking phase. After the whole braking cycle was completed and the assumed maximum initial speed was overcome, the supercapacitor reached a maximum value of 2-3 volts. This was expected because

the mechanical energy from the small wheel and motor assembly cannot provide the same amount of energy as a supercapacitor.

The results provided accurate predictions based on the assumptions that have been made. The results suggest that energy gets transferred into the supercapacitor from the rotating motor via regenerative braking and that how quickly the supercapacitor charges is also in line with what would be predicted by the relationships between the speed of rotation, voltage generated, and energy charged into a supercapacitor. The results indicate that the energy recaptured at small scale will be modest and consistent with laboratory testing.

The early results give indication that the engineering drawings, along with the control schemes, can successfully show some regenerative brake functionality. The follow-up task will consist of constructing and testing specially designed motor controller hardware and the ability to interface that motor controller with the rest of the electronics (microcontroller, motor, and supercapacitor). The data obtained from testing will then be compared against the simulated data for purposes of verifying how accurate the model is, along with determining the actual efficiency of the regenerative braking system in real time.

While the analytical calculations and simulations demonstrate that regenerative braking is theoretically feasible within the designed system, practical implementation introduces additional constraints. In particular, the reduced supply voltage, and limitations in gate drive capability. These factors limit the ability of the motor to reach sufficient speed and prevent proper switching of the MOSFETs, thereby restricting the conditions required for regenerative braking. As a result, although the design is valid from a theoretical standpoint, hardware limitations must be addressed to achieve full system functionality.

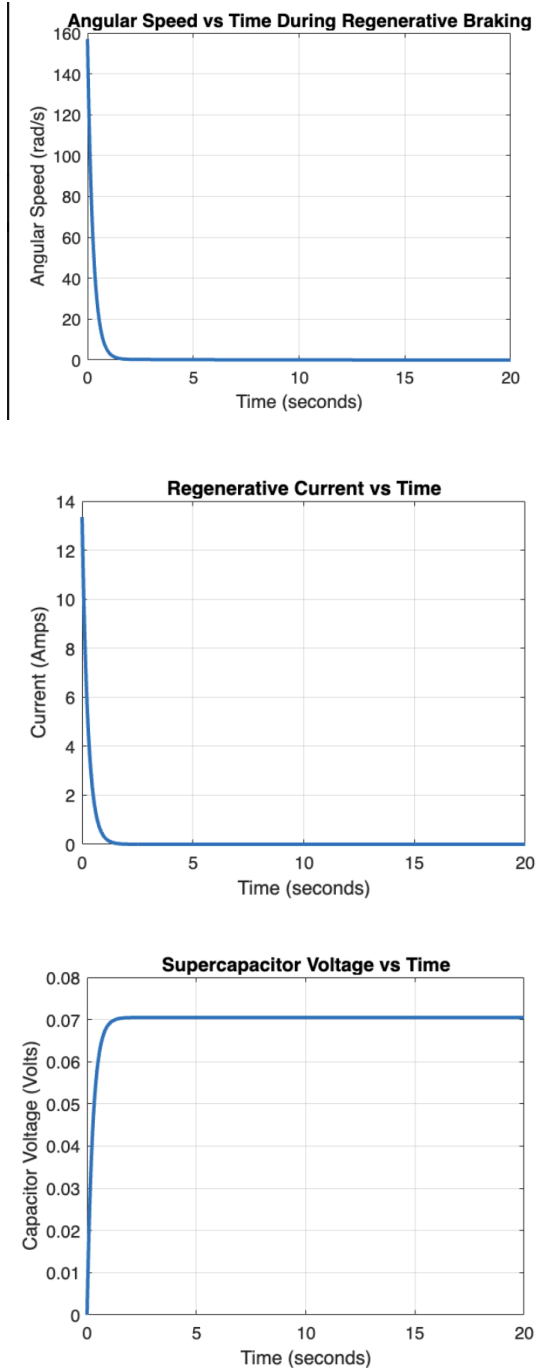


Figure [3,4,5] These are the simulations conducted using MATLAB, the MATLAB file is attached with the submission and code is provided below.

4 Results and Validation

4.1 Prototype

The proposed prototype consists of several key components, including a brushless motor, a DC power supply (implemented using a wall adapter), a three-phase inverter/rectifier, an ESP32 microcontroller interfaced with 3 L6387ED gate drivers ICs, 6 TPH2R306NH MOSFETs, 10F, 2.7V supercapacitor for energy storage, external 10V power supply for the gate drivers, and an energy recovery stage composed of a buck converter.

During normal operation, DC power from the supply is delivered to the three-phase inverter. The inverter is controlled by the ESP32, which generates pulse-width modulation (PWM) signals to switch the MOSFETs between high and low states through the gate drivers. By appropriately sequencing these switching signals, the inverter converts the DC input into three-phase AC, allowing the motor to operate.

To demonstrate regenerative braking, the switching sequence of the MOSFETs is modified through software code. In this mode, the motor operates as a generator, converting mechanical energy back into electrical energy. The generated three-phase AC is then routed through the inverter, which now functions as a rectifier, converting the AC back into DC. This recovered energy is transferred through a DC bus and subsequently passed into the buck converter, where the voltage is regulated to a suitable level for safe storage in the supercapacitor.

The DC bus is incorporated as an intermediate energy transfer stage within the system. Its primary function is to stabilize the DC voltage and reduce ripple effects caused by high-frequency switching in the inverter. By maintaining a more consistent voltage level, the DC bus

improves overall system performance and enables efficient energy transfer during both motoring and regenerative braking modes.

For development and testing, the inverter will initially be constructed using three separate breadboards, with each board representing an individual phase. This approach allows for independent verification of MOSFET switching behavior, including proper high-side and low-side operation, before integrating the complete three-phase system. Additionally, this method reduces the risk of wiring errors during the building stage.

The selection of the ESP32 microcontroller is based on its capability to generate multiple high-resolution PWM signals required for controlling the three-phase inverter and the buck converter. Its ample number of PWM pins make it well-suited for handling the timing, control, and interfacing requirements of the system.

Finally, the inductor used in the buck converter will be built by winding conductive wire around a cylindrical core. This approach allows the inductance value to be tailored based on the number of turns and physical dimensions, ensuring compatibility with the desired operating conditions of the converter.

3 Phase Inverter components

Name	Cost per 1	Description	How many units?	Total Unit	Cost total	Link to website
Input RC/Logic Filter 100 pF Capacitor	\$0.10000	CAP CER 100pF 25V X7R 0402	2 x 3 x2	12	\$1.20	https://www.digikey.co.uk/en/products/detail/w%C3%BCrth-elektronik/885012205038/5453388?srltid=AfmBOor76QAJNVgDlx6o4IdGqige0k_zC9yyAy8XtgrK6zoXeuoCvG
Driver local decoupling Capacitor	\$0.12000	CAP CER 0.47uF 25V X7R 0603	1 x 3 x2	6	\$0.72	https://www.digikey.ca/en/products/detail/w%C3%BCrth-elektronik/885012206075/5453848

470nF or 0.47uF						
L6387ED Half Bridge Gate Driver IC	\$2.50000	IC DRIVER HI/LO SIDE HV 8-SOIC	1 x 3 x2	6	\$15	https://www.digikey.ca/en/products/detail/stmicroelectronics/L6387ED/1665597
Boot Strap/ High Side Diode STPS0560Z	\$0.18000	DIODE SCHOTTKY 60V 500MA SOD123	1 x 3 x2 +1	7	\$1.26	https://www.digikey.ca/en/products/detail/stmicroelectronics/STPS0560Z/691885
Bootstrap 1uF capacitor	\$0.11000	CAP CER 1uF 25V X7R 0603	1 x 3 x2	6	\$0.66	https://www.digikey.ca/en/products/detail/w%C3%BCrth-elektronik/885012206076/5453849
Gate Resistor 33 ohms	\$0.01751	RES SMD 33 OHM 1% 1/10W 0603	2 x 3 x2	12	\$2.10	https://www.mouser.ca/ProductDetail/Vishay/CRCW060333R0FKEA?qs=ZDyrRr7H7mlo1WK7MqD7IA%3D%3D&srltid=AfmBOooAAvh3AZlv2l83DiGPAK9Ymlo-jVjdHLwHPBu_UfCyUKNCqEhN
Gate pull down resistor 10k ohms	\$0.10000	RES SMD 10K OHM 1% 1/16W 0402	2 x 3	6	\$0.6	https://www.digikey.ca/en/products/detail/yageo/RC0402FR-0710KL/726523
Power Mosfets TPH2R306NH	\$2.50	N-channel 60 V, 1.9 mO typ, 120 A STripFET™ F7	2 x 3 x3	18	\$45	https://www.digikey.ca/en/products/detail/toshiba-semiconductor-and-storage/TPH2R306NH-L1Q/3945692
Shunt Filter Resistor 1 ohm	\$0.10000	RES SMD 1 OHM 5% 1/8W 0805	1 x 3 x2	6	\$0.6	https://www.digikey.ca/en/products/detail/panasonic-electronic-components/ERJ-6GEYJ1R0V/282154
Shunt Sense RC Filter Capacitor 10nF or 10000pF	\$0.10000	CAP CER 10000PF 100V X7R 0603	1 x 3 x2	6	\$0.6	https://www.digikey.ca/en/products/detail/panasonic-electronic-components/ERJ-6GEYJ1R0V/282154

Total Cost	\$67.68
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Hall sensor development components

Name	Cost per unit	How many Units	Total Unit	Cost total	Link
100nF or 0.1μF Capacitors	\$0.08000	4 x2	8	\$0.64	https://www.digikey.ca/en/products/detail/yageo/CC0603KRX7R9BB104/2103082
3.3k ohm Resistors	\$0.10000	3	3	\$0.3	https://www.digikey.ca/en/products/detail/yageo/RC0603FR-073K3L/727126
2.2k ohm Resistors	\$0.10000	3 x2	6	\$0.6	https://www.digikey.ca/en/products/detail/yageo/RC0603FR-072K2L/727016
N4148W Diode	\$0.11000	1 x2	2	\$0.22	https://www.digikey.ca/en/products/detail/diodes-incorporated/1N4148W-7-F/814371

Total Cost	\$1.76
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Parts pre-owned:

- Microcontroller
- Potentiometer
- Pushbutton
- Jumper wires
- Inductor wire
- Breadboard

Table for Pins

Function	Type	ESP 32 Pin
Inverter PWM – Phase U High	PWM out	GPIO 25
Inverter PWM – Phase U Low	PWM out	GPIO 26
Inverter PWM – Phase V High	PWM out	GPIO 27
Inverter PWM – Phase V Low	PWM out	GPIO 14
Inverter PWM – Phase W High	PWM out	GPIO 12
Inverter PWM – Phase W Low	PWM out	GPIO 13
Buck PWM - High	PWM out	GPIO 32
Buck PWM - Low	PWM out	GPIO 33
Potentiometer (speed command)	ADC in	GPIO 34 (ADC1_CH6)
Shunt Current Sense 1	ADC in	GPIO 35 (ADC1_CH7)
Shunt Current Sense 2	ADC in	GPIO 36 (ADC1_CH0)
Shunt Current Sense 3	ADC in	GPIO 39 (ADC1_CH3)
Hall Sensor 1	Digital in (interrupt)	GPIO 4
Hall Sensor 2	Digital in (interrupt)	GPIO 16

Hall Sensor 3	Digital in (interrupt)	GPIO 17
Brake Button	Digital in	GPIO 18
Emergency Kill Switch	Digital out	GPIO 19

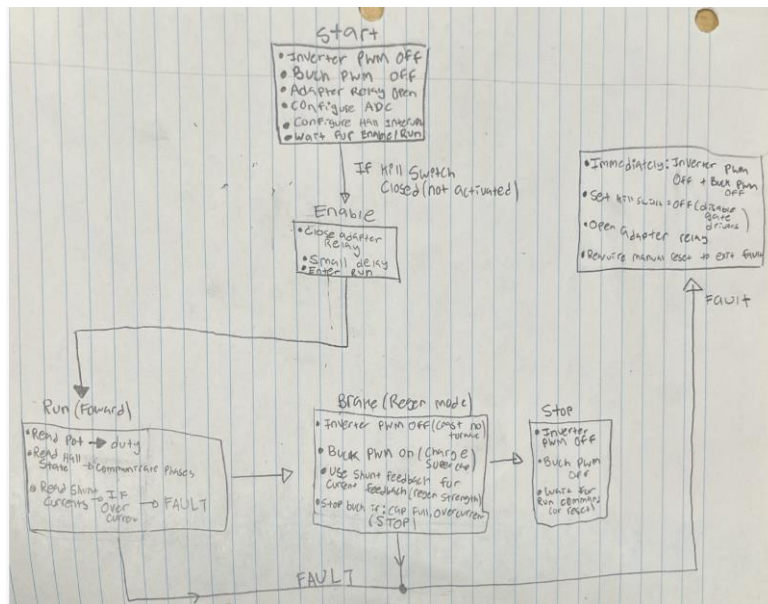


Figure [6] Control Flow Diagram

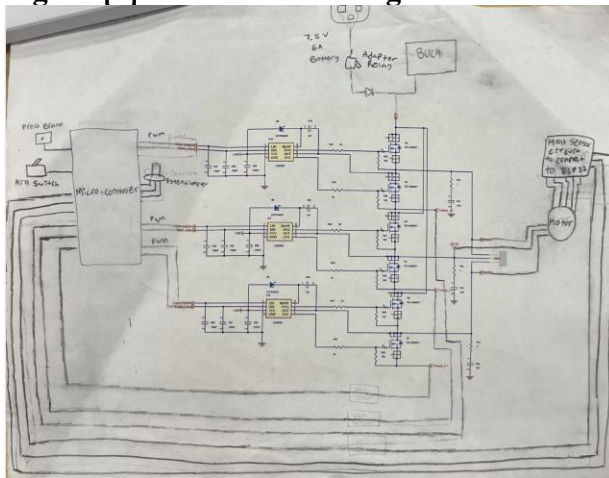


Figure [7]: Circuit diagram for Inverter and Microcontroller

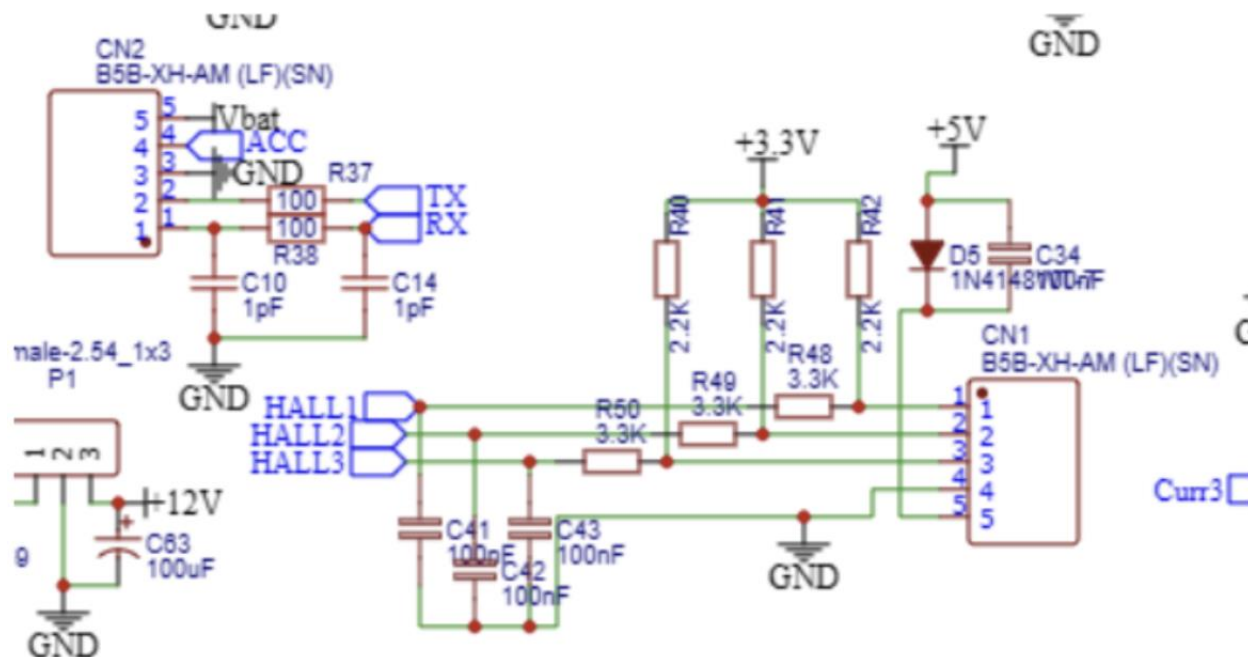


Figure [8]: Diagram for Hall Sensor to connect to ESP32

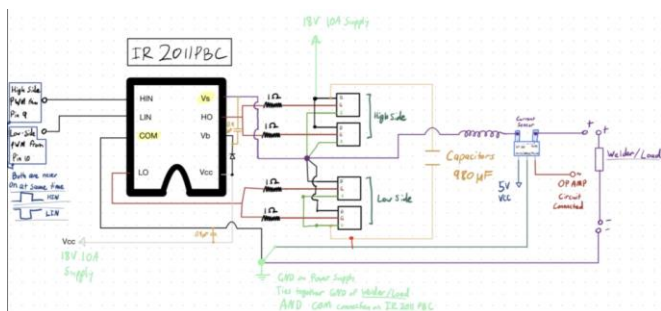


Figure [9]: Circuit diagram for Buck Converter

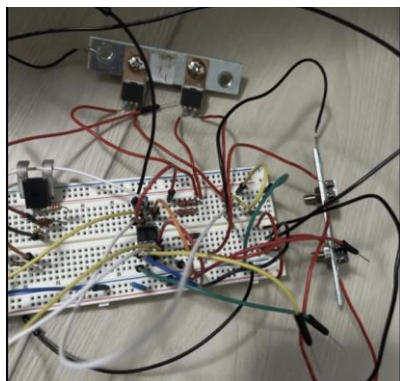


Figure [10]: Physical implenation of overall buck converter



Figure [11]: PWM signal for Buck Converter

Control Strategy: Six-Step Commutation

Six-step commutation energizes two motor phases at a time (one phase connected to the positive rail and one phase connected to the negative rail), while the third phase is left floating. Each commutation state is held until the Hall sensor pattern changes, indicating that the rotor has advanced to the next 60° electrical sector. By cycling through six states per electrical revolution, the controller produces a rotating stator field that pulls the rotor forward. In practice, the microcontroller drives the three gate-driver ICs (one per phase leg) to switch the appropriate high-side and low-side MOSFETs for each state.

State	High-side ON	Low-side ON	Floating phase
1	U	V	W
2	U	W	V
3	V	W	U
4	V	U	W
5	W	U	V
6	W	V	U

Example six-step commutation sequence (phase labels U/V/W).

During braking, motoring commutation PWM is disabled so no positive torque is commanded. The spinning wheel then back-drives the motor, generating three-phase voltage. The inverter's body diodes provide a rectification path back to the DC bus. The buck converter is

enabled to draw controlled current from the bus, producing generator current and an opposing electromagnetic torque.

4.2 Validation/Testing Strategy or Protocols

To verify that the regenerative braking system operates safely and meets the defined performance requirements, a subsystem-based testing approach is implemented. The system consists of a BLDC motor, a custom motor controller, an ESP32 microcontroller for commutation and control, a supercapacitor for energy storage, and a buck converter for voltage regulation. Each subsystem is validated independently prior to full system integration to reduce the risk of component damage and to facilitate early fault detection.

Initial testing focuses on the motor controller, specifically the MOSFET switching and gate driver circuits. The controller uses a three-phase inverter composed of three half-bridges, each driven by a dedicated gate driver IC to control the corresponding motor phase. To simplify debugging and minimize risk, each half-bridge is tested independently on a separate breadboard. This allows verification of proper high-side and low-side switching behavior before full integration. Oscilloscope measurements are used to observe the gate signals and output switching of the MOSFETs. These measurements ensure that the gate drivers respond correctly to the PWM control signals and confirm the absence of unintended switching events.

The next step is verifying the microcontroller's PWM output and control signals. The microcontroller outputs PWM signals to control the gate drivers for the MOSFETs in the motor controller, which will determine when each motor phase is switched during both normal and regenerative braking operations. Using an oscilloscope, the frequency, duty cycle, and timing relationship between each phase can be compared to verify that the signals are being produced

correctly. An important component to verify is the dead time between the high-side and low-side switching signals, as this ensures that no double-on current flows through either of the MOSFETs in the same half-bridge, which could create a risk of shortening the power supply.

After checking the microcontroller PWM signals, the next step will involve testing the buck converter subsystem to ensure it regulates the generated voltage to a level suitable for charging the supercapacitor. Initial testing is performed at low current to confirm stable operation and consistent output voltage, followed by testing under increasing load conditions using appropriately rated gauge wire to safely carry expected load currents.

System-level testing begins by operating the motor with the custom-built controller using a low-voltage power supply. For this testing phase, the motor will use a 7.5V laptop power supply for the initial power source. The motor will be powered on at a low speed and increase after it performs as expected. Powering on the motor at low speed will help eliminate the risk of damaging either the controller or motor if any unexpected behavior occurs. Hall effect sensor signals are monitored to verify correct rotor position detection and commutation. Motor speed is validated through both software readings and sensor signal frequency. Current sensing is verified by measuring the current and voltage values across the motor controllers shunt resistors and comparing against expected values. Obtaining these measurements is essential because they will allow the control software to monitor the operation of the motor and will prevent too much current from flowing, as this could damage the controller and the supercapacitor.

The regenerative braking testing will take place after stable motor operation is achieved. This test consists of first using the external power supply to accelerate the motor to a medium speed and then switching into regenerative brake mode so that the switching sequence of the MOSFETs will allow the motor to be used as a generator, converting mechanical energy from

the wheel into electrical energy. The generated energy will then pass through the buck-converter to be stored in the supercapacitor. During the regenerative test, various system parameters will be recorded, such as the angular velocity of the motor, the regenerative current measured through shunt resistors, and the voltage across the supercapacitor. The team will analyze the measurements taken during the test to determine if there has been a successful transfer of energy from the motor to the storage system.

The testing procedures outlined in previous sections will incorporate safety measures. An emergency shutoff, such as a circuit breaker or emergency stop switch, will be included in the power path so that the system can immediately disconnect from the input in the event a fault occurs. All initial tests will be performed at reduced speed and current so the amount of risk can be reduced. While testing, system parameters will be continuously monitored for parameters such as capacitor voltage and motor current, ensuring the component limits are not exceeded.

Because the testing protocols are structured, each sub-system can be validated independently through the testing process until the entire system is assembled in an integrated way. Verification of the PWM signals, gate driver circuits, buck converter operation, motor control functional operation, and regenerative braking operation will allow the team to have an adequate level of confidence that the fully integrated system meets the original design specifications.

4.3 Final Test Results

Full system operation was not achieved during the testing phase. The BLDC motor did not run under the custom controller, which meant regenerative braking could not be

experimentally demonstrated and no direct measurements of current, voltage, or energy recovery were obtained.

Although the complete system could not be tested as a whole, several key subsystems were successfully validated in isolation. The validation focused on confirming that the electrical design, including the microcontroller outputs, Hall-sensor interface, gate-driver inputs, and MOSFET gate behavior, operated as intended prior to full motor integration. Continuity checks verified correct power and ground routing across the breadboard and ESP32 connections, and inspection and testing confirmed no unintended shorts across solder joints and wiring for the SMT components.

Using an oscilloscope, PWM waveforms were measured at the gate-driver inputs and shown to have the expected frequency, duty cycle, and complementary timing relationships. A timing inconsistency initially observed with the selected PWM peripheral was found and determined to be based on the ESP32 pins using different internal timers. This issue was resolved by migrating away from the unstable timer configuration by changing pin values, after which all three gate-driver channels exhibited the expected high-side input behavior.

During integration testing, it was also observed that the ICs could not pull the high side PWMs due to the ESP32's 3.3 V output consistently, thus we switched to using the 5 V Arduino output; switching to a 5 V logic-level input improved the observable switching behavior at the driver interface, indicating that logic-level compatibility and noise margin were limiting factors in the initial configuration.

These results confirm that in isolation, these key components of the system did function as intended. A detailed analysis identified several compounding factors that prevented full system operation, the first being that the measured gate-to-source voltages across the MOSFETs

were approximately 2 V which were significantly below the ~ 10 V threshold required for proper switching. Although high-side PWM signals were successfully generated and delivered to the gate-driver ICs, only 2 of the 6 MOSFETs consistently showed measurable gate-to-source drive voltage, preventing the full three-phase commutation and therefore preventing sustained motor rotation and regenerative braking demonstration within the project timeline.

The most critical hardware constraint was the breadboard's inability to handle the 6 A of current required during operation. The original plan was to prototype on a breadboard before transitioning to a PCB; however, the breadboard's current limitations were not adequately accounted for during planning. The resulting voltage drops and unstable operation under load were the primary suspected cause of the gate drive failures observed across the majority of MOSFETs.

Beyond the breadboard itself, the overall power supply and wiring configuration were insufficient to deliver consistent voltage to the inverter under load conditions. Long wire lengths also introduced parasitic inductance and resistance that negatively impacted switching performance, compounding the instability caused by the breadboard limitations.

It is also worth noting that several of the challenges encountered were consequences of design choices made earlier in the project such as the use of SMT components, which proved difficult to solder reliably, and the initial incorrect pin assignments on the ESP32, which were subsequently corrected.

Simulation and theoretical analysis had predicted recoverable energy of 2–5 J per braking event, an observable voltage increase across the supercapacitor during braking, and stable regenerative current flow. However, due to the hardware limitations described above, these experimental results could not be experimentally validated.

While full system validation was not achieved within the project timeline, the work accomplished confirms the design's feasibility at a subsystem level. PWM generation and timing were quantitatively validated, wiring integrity was verified, and logic-level compatibility issues were identified and resolved. Simulation results further support the theoretical viability of the regenerative braking approach.

Future implementation should prioritize migrating from the breadboard to a PCB or perfboard rated for the required current and voltage levels. This change alone is expected to resolve the primary failure mode observed and enable the full three-phase commutation and energy recovery behavior that the design targets. These results showcase the challenges of implementing high-current power electronic systems using prototyping platforms and highlight the importance of proper hardware design considerations in achieving reliable system operation.



Figure [12] PWM generation

5 Project Team Management

5.1 Team Member Task Breakdown and contributions and timelines

To guarantee a balanced workload, technical depth, and effective system integration, the project is managed by a three-person team. To facilitate shared problem-solving and integration, each team member is given the primary control of a significant subsystem while preserving overlap.

The first team member primary responsibilities are Hardware integration and experimental setup. The BLDC motor, motor controller, energy storage parts, and protective circuits must all be chosen and physically integrated. In addition, this person is responsible for the bench-scale test rig's design and construction, which guarantees reliable testing conditions, safe electrical connections, and mechanical stability. Sensor placement, wiring verification, and supporting experimental testing and troubleshooting are additional duties.

The second team member concentrates on motor controller configuration and embedded control. Setting current and voltage restrictions, implementing braking control logic using an integrated microcontroller, and configuring the BLDC motor controller for motoring and regenerative braking operation are among the duties. Additionally, this component facilitates regulated switches between drive and regenerative modes by controlling communication between the microcontroller and motor controller. Assisting with data collecting, maintaining safe operation during braking events, and adjusting control parameters are some support tasks.

The third team member oversees modelling, simulation, and data analysis. This entails building analytical and simulation-based models of the BLDC motor and regenerative braking system using MATLAB/Simulink. By analyzing experimental data collected during testing,

system performance is evaluated, including recovered energy, brake torque, and efficiency trends. Additionally, this individual helps with technical documentation and reporting and aids validation by contrasting experimental results with theoretical expectations.

To guarantee a coherent and well-understood design across all subsystems, the team maintains shared engagement in system integration, testing, documentation, and final analysis despite clearly defined duties.

5.2 Team Communication and Organization Methodology

Maintaining efficient coordination and communication requires an organized yet adaptable strategy. The team uses a mix of in-person discussions and online communication platforms to organize work, share progress reports, and resolve technical issues. Design notes, simulation results, experimental data, and draft report sections are stored in shared online documentation to guarantee version consistency and team member accessibility.

Milestone-based checklists that are in line with the project timeline are used to arrange task assignments and short-term objectives. This method keeps system-level integration needs visible while enabling the team to monitor progress at the subsystem level. To guarantee agreement and lower the possibility of design discrepancies, technical decisions are considered jointly.

5.3 Meetings/Attendance Listing

The team maintained consistent communication throughout the project using a combination of in-person meetings and digital communication tools. Weekly meetings were held to review progress, assign tasks, and address technical challenges.

Communication strategies included:

- Regular progress meetings: Weekly team meetings to track milestones and identify issues
- Task coordination: Division of responsibilities to allow parallel work on hardware, control, and analysis
- Iterative problem-solving: Collaborative debugging sessions to identify and resolve technical issues
- Advisor consultations: Biweekly meetings with the faculty advisor during the second semester to review progress and receive guidance

This structured approach allowed the team to maintain steady progress and adapt to challenges. However, the complexity of the hardware implementation required more iterative testing than initially anticipated.

One limitation in team organization was the decision to integrate all subsystems simultaneously rather than validating each stage incrementally. This made debugging more difficult and delayed identification of key issues.

Date	Time	Duration	Description
Sept 14, 2025	12:00 PM	1 hour	Project scope, objectives, and system concept
Sept 26, 2025	12:00 PM	1 hour	Component selection and budget constraints
Oct 12, 2025	1:00 PM	2 hours	Literature review findings and system Architecture “Project Proposal”
Oct 26, 2025	2:00 PM	1 hour	Preliminary analysis and regenerative modeling
Nov 9, 2025	12:00 PM	1 hour 30 mins	Documentation and design review preparation “Design Review 1”
Nov 23, 2025	12:00 PM	1 hour	Midterm review report
Dec 7, 2025	1:00 PM	2 hours	“Midterm Review Report”
Jan 14, 2026	3:45 PM	30 mins	Meeting with advisor (Brown), getting feedback
Jan 22, 2026	1:00 PM	45 mins	Ordering parts
Jan 27, 2026	3:45 PM	45 mins	Meeting with advisor (Brown), getting

			feedback
Feb 4, 2026	2:00 PM	1 hour	Design review 2/ building system
Feb 12, 2026	3:45 PM	30 mins	Meeting with advisor (Brown), getting feedback
Feb 25, 2026	2:00 PM	30 mins	“Electrical Requirements” and “Design Review 2”
Mar 5, 2026	3:45 PM	2 hour 15 mins	Meeting with advisor (Brown), getting feedback/ building system/ “Design Validation and Test Plan”
Mar 19, 2026	3:45 PM	3 hour 30 mins	Meeting with advisor (Brown), getting feedback/ building system/ “Design Showcase Abstract”
Mar 24, 2026	10:00 AM	6 hours	Testing and validation of full system
Mar 25, 2026	10:00 AM	7 hours	Testing and validation of full system
Mar 26, 2026	9:00 AM	8 hours	Testing and validation of full system
Apr 2, 2026	4:00 PM	2 hours	Final report
Apr 8, 2026	2:00 PM	3 hours	Final report
Apr 9, 2026	12:00 PM	5 hours	“Final Report”

6 Final Conclusions

This project aimed to design and showcase a low-cost regenerative braking system with a custom designed three phase motor controller. The system consisted of a three-phase inverter, microcontroller-based control, and a supercapacitor energy storage with the end goal to be able to show energy recovery during braking.

The result showed that, while the system was successfully built and partially validated at a subsystem level, full system integration was not achieved. Key components, including PWM signal generation and gate driver operation were verified through experimental testing. However, due to limitations in hardware implementation such as insufficient gate drive voltage, breadboard current constraints, and power delivery problems, prevented the successful motor operation which prevented the showcasing of regenerative braking.

However, despite the challenges faced, the project provided valuable insight into the design and implementation of power electronics systems. Theoretical analysis and simulation confirmed the feasibility of regenerative braking within the designed system, with the predicted recovery in energy in the range of 2-5 J per braking event.

Overall, the project met the design objectives in terms of system development, analysis, and identification of critical implementation challenges. The result highlights the importance of proper hardware design and integration when developing custom motor control systems.

7 Future Work and Recommendations

For the next iteration of this project, the primary improvement should be transitioning from a breadboard-based build to a PCB or a perfboard layout with short, high-current bus wiring. This will improve signal integrity, reduce noise and parasitic inductance, improve current handling, ensure reliable electrical connection, and improve overall circuit reliability issues that are likely to contribute to the inability to achieve stable motor operation in the current prototype.

It's important to insure that optimizing the gate drive circuitry would allow sufficient voltage for MOSFET switching. Reducing wire length would minimize parasitic inductance and resistance. Future work should avoid integrating all subsystems at once, where individual subsystems are fully tested and validated before integration. In addition, the power system should be reevaluated, as the present supply voltage and/or available current may be limiting motor startup and commutation performance thus using a supply closer to the motor/controller operating range would increase the likelihood of achieving consistent results.

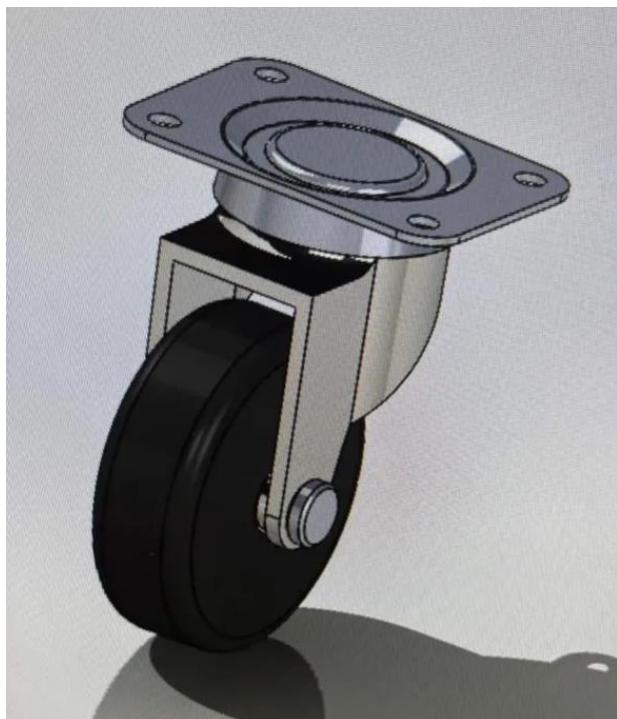
The control system can also be strengthened by implementing a more complete closed-loop approach, using Hall feedback for commutation and speed estimation alongside improved current feedback for protection and regulation. To support this, the shunt measurement circuitry should be upgraded by adding a current-sense amplifier or op-amp stage and improved filtering so that current readings are stable and meaningful for both protection and data collection. On the energy recovery side, the buck converter and supercapacitor subsystem should be validated independently before full integration: the buck should be tested on a bench supply as a standalone charger, the inductor should be built or selected to meet the target value of $47\ \mu\text{H}$ with adequate current rating, and finally, the supercapacitor should be characterized using scope-based charge/discharge measurements to confirm real capacitance.

8 References

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9 Appendices



Test Rig to support Motor, with intention to screw onto flat platform having motor face vertically

Untitled Gantt Project

Apr. 9, 2026

<http://>

Project manager	Damanpreet, Chad , Sufyan, All Group Members
Project dates	Oct. 20, 2025 - Apr. 21, 2026
Completion	62%
Tasks	41
Resources	4

Untitled Gantt Project

Apr. 9, 2026

Tasks

2

Name	Begin date	End date
Planning & Management	2025-10-20	2026-04-20
Project kickoff, roles, success metrics, safety rules	2025-10-20	2025-10-24
Lab safety plan & SOP's (battery, ESD, high current, etc)	2025-10-29	2025-11-05
Confirm system requirements (voltage range, temp limits)	2025-10-29	2025-11-06
System Architecture & interfaces(power, sensing, logging)	2025-11-03	2025-11-28
Sign off vendors/part numbers	2025-11-12	2026-01-02
Demo rehearsal & dry-runs (script, timing, contingency)	2026-04-01	2026-04-09
Contingency buffer (hardware fixes, final tuning)	2026-04-10	2026-04-17
Final demo	2026-04-20	2026-04-20
Hardware: Rig, Power, Integration	2025-11-05	2026-04-03
Select BOM (DD BLDC, VESC 6+, Li-ion pack, sensors, wiring)	2025-11-05	2025-11-11
Procure long-lead parts	2025-12-30	2026-01-27
Design rig mechanics (base, mounts, guards)	2026-01-12	2026-02-10
Power wiring & protection plan (fuses, shunts, connectors)	2026-01-19	2026-02-20
Bench integrate battery pack (basic charge/discharge tests)	2026-01-20	2026-03-09
Fabricate & assemble rig v0 (wheel/flywheel, brackets, guards)	2026-01-21	2026-03-09
Thermal checks & airflow tweaks (stay <100 °C; add fan/duct if needed)	2026-03-05	2026-03-24
Demo rig polish (enclosure, labels, cable management, guards)	2026-03-10	2026-04-03
Firmware & Control (ESP32 + VESC)	2025-12-10	2026-03-24
Set up ESP32 toolchain & repo (PlatformIO/Arduino)	2025-12-24	2026-01-19
Define telemetry schema (V, I, RPM, T; sample rates/IDs)	2025-12-29	2026-01-22
Bring up VESC comms (UART/CAN; read RPM/V/I)	2025-12-10	2026-01-30
IMU + hall sensor drivers (calibration, filtering)	2025-12-15	2026-02-04
Control flow diagram (Idle, Drive, Brake-Regen, Fault)	2025-12-22	2026-01-27
Controller choice study (PI/PD/PID recommendation)	2026-01-07	2026-01-13
Open-loop regen command path (negative current step tests)	2026-01-14	2026-01-27
DAQ to Serial Studio/BLE dashboard (real-time plots + logging)	2026-01-07	2026-01-30
Closed-loop current/torque PI tuning (anti-windup, limits)	2026-01-28	2026-02-10
Decel/comfort PD & ramp shaping (jerk limits, rate limiters)	2026-02-11	2026-02-17

Untitled Gantt Project

Apr. 9, 2026

Tasks

3

Name	Begin date	End date
Log replay & parameter sweeps (batch tests via scripts)	2026-03-02	2026-03-09
User guide & safety quickstart (startup, limits, faults)	2026-03-16	2026-03-24
Modeling & Data Analysis	2025-11-12	2026-03-19
Build MATLAB/Simulink model (Ke, Kt, Ra, La; VESC limits)	2025-11-12	2026-01-15
Testbench parameter ID (back-EMF, inertia estimate)	2025-12-10	2026-01-23
Simulate control law options (stability, overshoot, settling)	2026-01-07	2026-02-10
Validate sim vs bench; tune gains/model updates	2026-02-02	2026-02-20
Results plots & tables (torque–speed, regen %, temps)	2026-03-10	2026-03-19
Testing & Validation	2026-02-16	2026-03-20
Test plan & matrix (RPM: 300–1200; decel targets; SOC bands)	2026-02-16	2026-02-20
Data runs (execute matrix; multiple trials per point)	2026-02-23	2026-03-13
Optimization pass (comfort/efficiency; gain scheduling vs RPM)	2026-03-02	2026-03-20

Untitled Gantt Project

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Resources

4

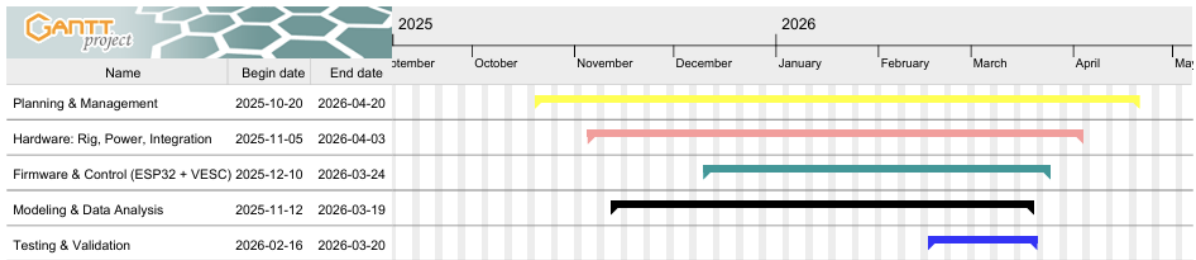
Name	Default role
Damanpreet	project manager
Chad	project manager
Sufyan	project manager
All Group Members	project manager

Untitled Gantt Project

Apr. 9, 2026

Gantt Chart

5



Untitled Gantt Project

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Resources Chart

6



[Capstone\(Final\).pdf](#)

TOSHIBA

TPH2R306NH

MOSFETs Silicon N-channel MOS (U-MOSTM-H)

TPH2R306NH

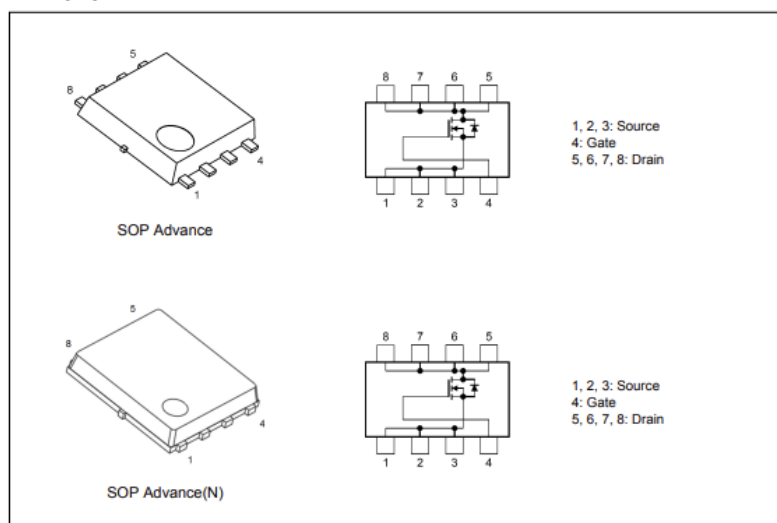
1. Applications

- DC-DC Converters
- Switching Voltage Regulators
- Motor Drivers

2. Features

- (1) High-speed switching
- (2) Small gate charge: $Q_{SW} = 26 \text{ nC}$ (typ.)
- (3) Low drain-source on-resistance: $R_{DS(ON)} = 1.9 \text{ m}\Omega$ (typ.) ($V_{GS} = 10 \text{ V}$)
- (4) Low leakage current: $I_{DSS} = 10 \text{ }\mu\text{A}$ (max) ($V_{DS} = 60 \text{ V}$)
- (5) Enhancement mode: $V_{th} = 2.0 \text{ to } 4.0 \text{ V}$ ($V_{DS} = 10 \text{ V}$, $I_D = 1.0 \text{ mA}$)

3. Packaging and Internal Circuit



The package can be selected according to your preference. For details, please contact your TOSHIBA sales representative.

Start of commercial production

2012-11

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Toshiba Electronic Devices & Storage Corporation

1

2019-10-24

Rev.3.0

https://toshiba.semicon-storage.com/info/TPH2R306NH_datasheet_en_20191024.pdf?did=13732&prodName=TPH2R306NH

3 Pin connection

Figure 2. Pin connection (top view)

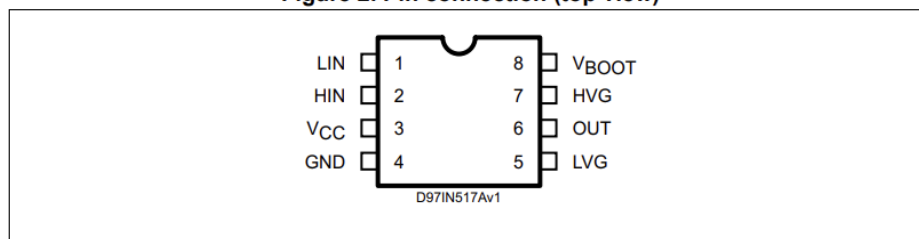


Table 4. Pin description

No.	Pin	Type	Function
1	LIN	I	Low-side driver logic input
2	HIN	I	High-side driver logic input
3	V _{CC}	P	Low voltage power supply
4	GND	P	Ground
5	LVG ⁽¹⁾	O	Low-side driver output
6	OUT	P	High-side driver floating reference
7	HVG ⁽¹⁾	O	High-side driver output
8	V _{BOOT}	P	Bootstrap supply voltage

1. The circuit guarantees 0.3 V maximum on the pin (at $I_{\text{sink}} = 10 \text{ mA}$). This allows to omit the "bleeder" resistor connected between the gate and the source of the external MOSFET normally used to hold the pin low.

<https://www.st.com/content/ccc/resource/technical/document/datasheet/09/06/7a/6f/1d/a6/45/f5/CD00173127.pdf/files/CD00173127.pdf/jcr:content/translations/en.CD00173127.pdf>

2.1 Pin Layout

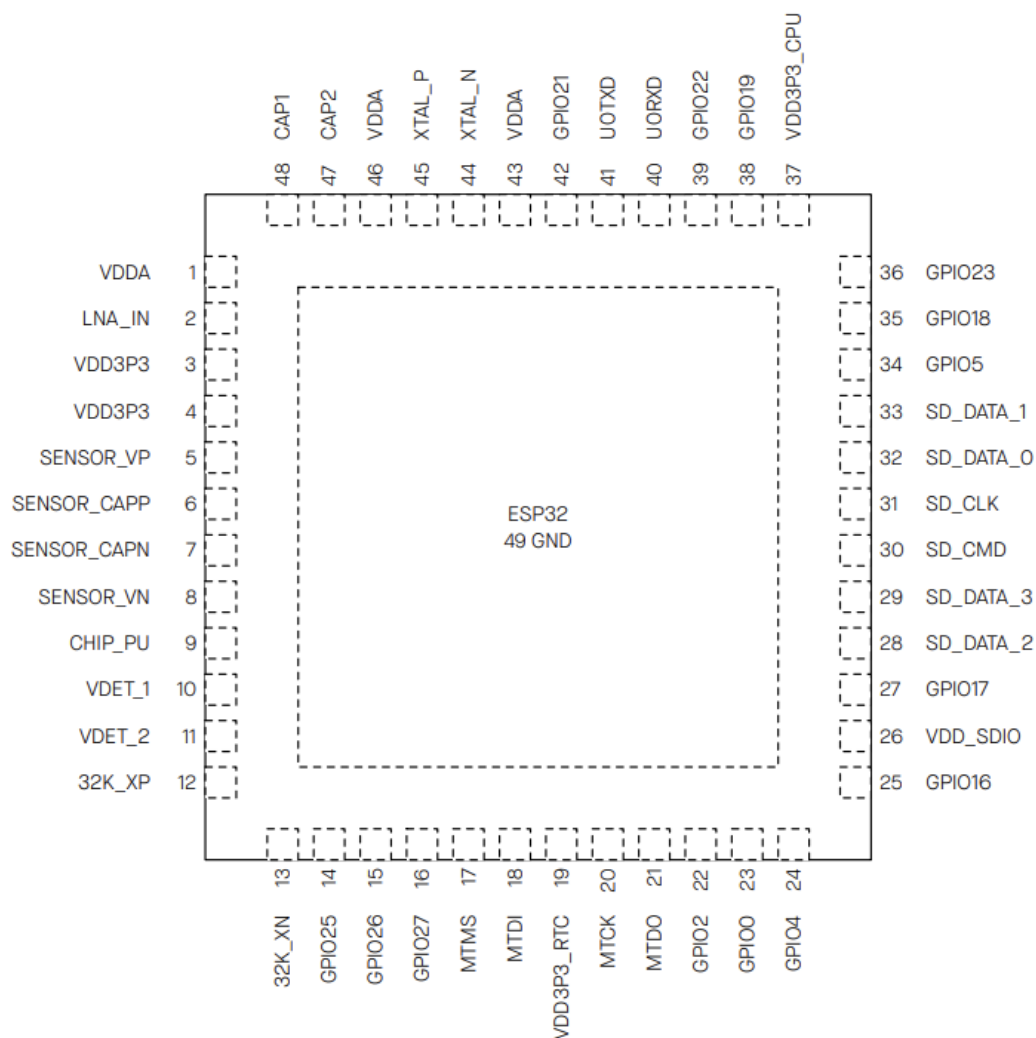


Figure 2-1. ESP32 Pin Layout (QFN 6*6, Top View)

https://documentation.espressif.com/esp32_datasheet_en.pdf



2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Max
	Conservative thermal limits for the whole board:	-40 °C (-40 °F)	85 °C (185 °F)

NOTE: In extreme temperatures, EEPROM, voltage regulator, and the crystal oscillator, might not work as expected.

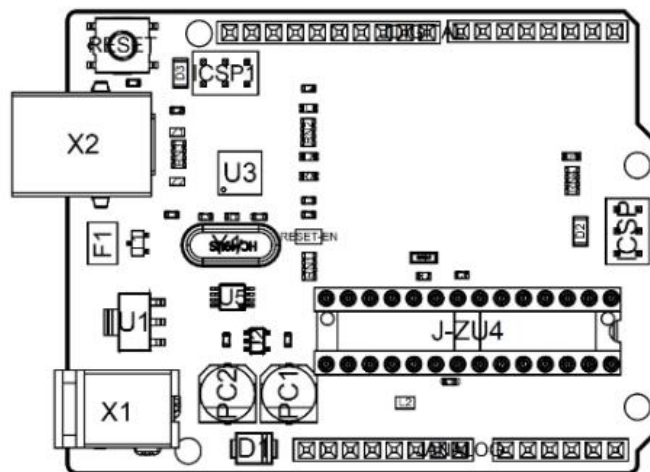
2.2 Power Consumption

Symbol	Description	Min	Typ	Max	Unit
VINMax	Maximum input voltage from VIN pad	6	-	20	V
VUSBMax	Maximum input voltage from USB connector		-	5.5	V
PMax	Maximum Power Consumption	-	-	xx	mA

3 Functional Overview

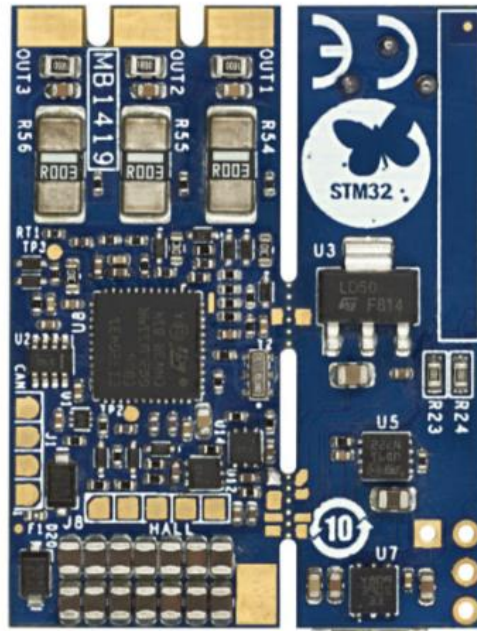
3.1 Board Topology

Top view



Board topology

**Figure 1. B-G431B-ESC1 Discovery kit
(front view)**

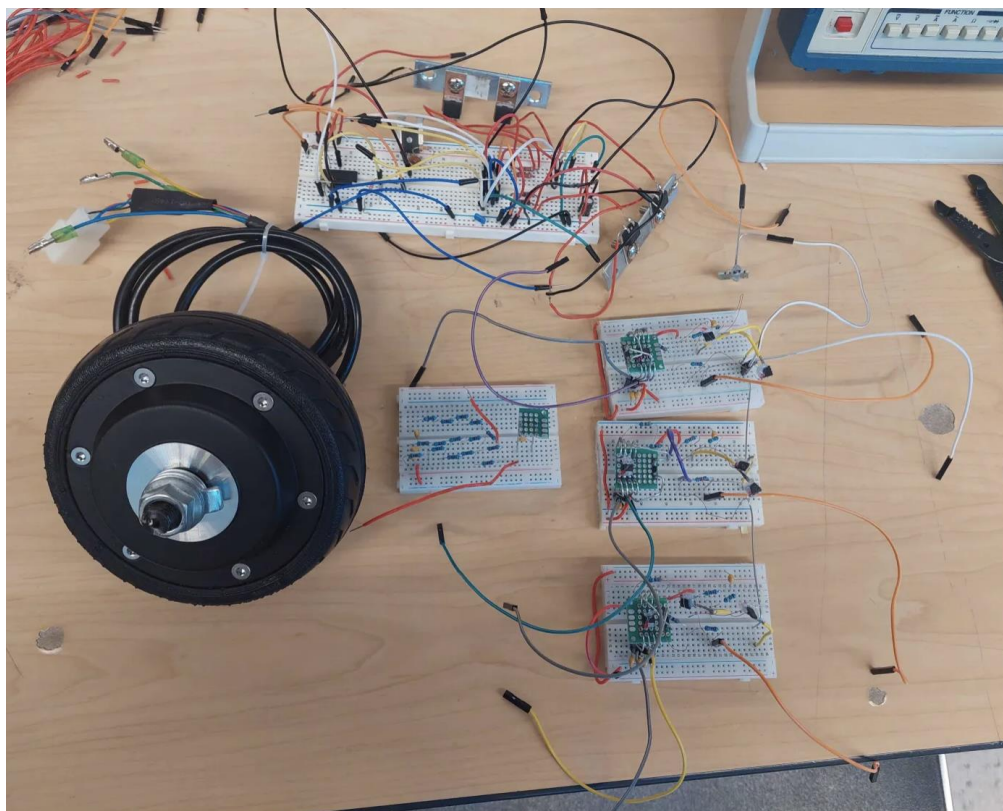


**Figure 2. B-G431B-ESC1 Discovery kit
(bottom view)**



Pictures are not contractual.

<https://www.digikey.ca/en/products/detail/stmicroelectronics/B-G431B-ESC1/10321670?s=N4lgTCBcDaIEIFoDiAWAZARkQUQMogEMQBdAXyA>



Arduino IDE code to just get motor to spin:

```
// =====
// BLDC Open-Loop Forced Commutation
// Arduino UNO Version
// =====

// --- Gate drive pins ---
const int AH = 9; // PWM
const int AL = 2;

const int BH = 10; // PWM
const int BL = 4;

const int CH = 11; // PWM
const int CL = 7;

// --- Ramp parameters ---
const int START_DELAY = 12000;
const int END_DELAY = 1500;
const int RAMP_STEPS = 600;
const int START_DUTY = 100;
const int END_DUTY = 180;
const int DEAD_TIME = 8;

// --- Runtime state ---
int step = 0;
int pwmDuty = START_DUTY;
long stepDelay = START_DELAY;
long rampCount = 0;
bool ramping = true;
unsigned long lastStepTime = 0;

// =====
// COMMUTATION TABLE
// =====
const int commSteps[6][6] = {
  {1, 0, 0, 0, 1, 0},
  {1, 0, 0, 0, 0, 1},
  {0, 1, 0, 0, 0, 1},
  {0, 1, 0, 1, 0, 0},
  {0, 0, 1, 1, 0, 0},
  {0, 0, 1, 0, 1, 0}
};

// =====
// SETUP
```

```

// =====
void setup() {
  Serial.begin(115200);

  pinMode(AL, OUTPUT); digitalWrite(AL, LOW);
  pinMode(BL, OUTPUT); digitalWrite(BL, LOW);
  pinMode(CL, OUTPUT); digitalWrite(CL, LOW);

  pinMode(AH, OUTPUT);
  pinMode(BH, OUTPUT);
  pinMode(CH, OUTPUT);

  allOff();

  Serial.println("Starting...");
  delay(500);
}

// =====
// ALL OFF
// =====
void allOff() {
  analogWrite(AH, 0);
  analogWrite(BH, 0);
  analogWrite(CH, 0);

  delayMicroseconds(DEAD_TIME);

  digitalWrite(AL, LOW);
  digitalWrite(BL, LOW);
  digitalWrite(CL, LOW);
}

// =====
// COMMUTATE
// =====
void commutate(int s) {
  const int* p = commSteps[s];

  // Turn OFF high sides
  analogWrite(AH, 0);
  analogWrite(BH, 0);
  analogWrite(CH, 0);

  delayMicroseconds(DEAD_TIME);
}

```

```

// Turn ON low sides
digitalWrite(AL, p[3]);
digitalWrite(BL, p[4]);
digitalWrite(CL, p[5]);

delayMicroseconds(1);

// Apply PWM to high sides
if (p[0]) analogWrite(AH, pwmDuty);
if (p[1]) analogWrite(BH, pwmDuty);
if (p[2]) analogWrite(CH, pwmDuty);
}

// =====
// RAMP UPDATE
// =====
void updateRamp() {
    if (!ramping) return;

    rampCount++;
    float t = (float)rampCount / RAMP_STEPS;

    if (t >= 1.0) {
        t = 1.0;
        ramping = false;
        Serial.println("Ramp complete.");
    }

    stepDelay = (long)(START_DELAY + t * (END_DELAY - START_DELAY));
    pwmDuty = (int)(START_DUTY + t * (END_DUTY - START_DUTY));
}

// =====
// LOOP
// =====
void loop() {
    unsigned long now = micros();

    if (now - lastStepTime >= (unsigned long)stepDelay) {
        lastStepTime = now;

        commutate(step);
        updateRamp();

        step++;
        if (step >= 6) step = 0;
    }
}

```

```
if (ramping && rampCount % 50 == 0) {  
  Serial.print("delay=");  
  Serial.print(stepDelay);  
  Serial.print(" duty=");  
  Serial.println(pwmDuty);  
}  
}  
}
```


Arduino IDE code for full system:

```
// =====
// BLDC Open-Loop Forced Commutation
// Arduino UNO Version
// =====

// This sketch drives a 3-phase Brushless DC (BLDC) motor in open-loop mode.
// "Open-loop" means there is no feedback from the motor (no Hall sensors or back-EMF
// sensing).
// Instead, the Arduino forces the commutation sequence at a timed interval,
// gradually speeding up (ramping) from a slow start to a target speed.

// --- Gate drive pins ---
const int AH = 3; // Phase A HIGH side — PWM pin (controlled by Timer2)
const int AL = 2; // Phase A LOW side — digital on/off only
const int BH = 5; // Phase B HIGH side — PWM pin (controlled by Timer0)
const int BL = 4; // Phase B LOW side — digital on/off only
const int CH = 6; // Phase C HIGH side — PWM pin (controlled by Timer0)
const int CL = 7; // Phase C LOW side — digital on/off only

// --- Buck PWM pins for Arduino UNO ---
const int BUCK_HIN = 9; // OC1A
const int BUCK_LIN = 10; // OC1B
const int MODE_BTN = 8; // push button to GND, using INPUT_PULLUP

// --- Ramp parameters ---
const int START_DELAY = 15000;
const int END_DELAY = 1500;
const int RAMP_STEPS = 600;
const int START_DUTY = 150;
const int END_DUTY = 220;
const int DEAD_TIME = 8;

// --- Runtime state ---
int step = 0;
int pwmDuty = START_DUTY;
long stepDelay = START_DELAY;
long rampCount = 0;
bool ramping = true;
unsigned long lastStepTime = 0;

// --- Mode state ---
bool buckMode = false;
bool lastBtnState = HIGH;

// =====
```

```

// COMMUTATION TABLE
// =====
const int commSteps[6][6] = {
    {1, 0, 0, 0, 1, 0}, // Step 0: AH on (PWM), BL on
    {1, 0, 0, 0, 0, 1}, // Step 1: AH on (PWM), CL on
    {0, 1, 0, 0, 0, 1}, // Step 2: BH on (PWM), CL on
    {0, 1, 0, 1, 0, 0}, // Step 3: BH on (PWM), AL on
    {0, 0, 1, 1, 0, 0}, // Step 4: CH on (PWM), AL on
    {0, 0, 1, 0, 1, 0}  // Step 5: CH on (PWM), BL on
};

// =====
// ALL OFF
// =====
void allOff() {
    analogWrite(AH, 0);
    analogWrite(BH, 0);
    analogWrite(CH, 0);

    delayMicroseconds(DEAD_TIME);

    digitalWrite(AL, LOW);
    digitalWrite(BL, LOW);
    digitalWrite(CL, LOW);
}

// =====
// BUCK PWM SETUP
// =====
void initBuckPwm() {
    pinMode(BUCK_HIN, OUTPUT);
    pinMode(BUCK_LIN, OUTPUT);

    digitalWrite(BUCK_HIN, LOW);
    digitalWrite(BUCK_LIN, LOW);

    // Timer1: Phase-Correct PWM, TOP = ICR1 (mode 10)
    // Outputs disconnected at init
    TCCR1A = _BV(WGM11);
    TCCR1B = _BV(WGM13) | _BV(CS10); // no prescaler
    TCNT1 = 0;

    ICR1 = 799; // 10 kHz phase-correct at 16 MHz
    OCR1A = 0;
    OCR1B = 0;
}

```

```

void stopBuckPwm() {
  // Disconnect OC1A / OC1B from Timer1, keep outputs low
  TCCR1A = _BV(WGM11);
  digitalWrite(BUCK_HIN, LOW);
  digitalWrite(BUCK_LIN, LOW);
}

void startBuckPwm() {
  // Turn inverter fully off first
  allOff();
  delayMicroseconds(DEAD_TIME);

  // Set duty values from your original example
  OCR1A = 80; // OC1A non-inverting
  OCR1B = 80; // OC1B inverting

  // Enable Timer1 complementary outputs
  TCCR1A = _BV(COM1A1) // OC1A non-inverting
    | _BV(COM1B1) // OC1B inverting
    | _BV(COM1B0)
    | _BV(WGM11);

  Serial.println("BLDC OFF -> Buck PWM ON");
}

// =====
// BUTTON CHECK
// =====
void checkModeButton() {
  bool btnState = digitalRead(MODE_BTN);

  // Detect button press once
  if (!buckMode && lastBtnState == HIGH && btnState == LOW) {
    allOff(); // turn off BLDC first
    delayMicroseconds(DEAD_TIME);
    buckMode = true;
    startBuckPwm(); // then start buck PWM
    delay(25); // simple debounce
  }

  lastBtnState = btnState;
}

// =====
// COMMUTATE

```

```

// =====
void commutate(int s) {
    const int* p = commSteps[s];

    // Turn OFF all high-side switches first
    analogWrite(AH, 0);
    analogWrite(BH, 0);
    analogWrite(CH, 0);
    delayMicroseconds(DEAD_TIME);

    // Set low-side switches
    digitalWrite(AL, p[3]);
    digitalWrite(BL, p[4]);
    digitalWrite(CL, p[5]);
    delayMicroseconds(1);

    // Apply PWM to correct high-side switch
    if (p[0]) analogWrite(AH, pwmDuty);
    if (p[1]) analogWrite(BH, pwmDuty);
    if (p[2]) analogWrite(CH, pwmDuty);
}

// =====
// RAMP UPDATE
// =====
void updateRamp() {
    if (!ramping) return;

    rampCount++;

    float t = (float)rampCount / RAMP_STEPS;

    if (t >= 1.0) {
        t = 1.0;
        ramping = false;
        Serial.println("Ramp complete.");
    }

    stepDelay = (long)(START_DELAY + t * (END_DELAY - START_DELAY));
    pwmDuty = (int)(START_DUTY + t * (END_DUTY - START_DUTY));
}

// =====
// SETUP
// =====
void setup() {

```

```

Serial.begin(115200);

// BLDC low-side pins
pinMode(AL, OUTPUT); digitalWrite(AL, LOW);
pinMode(BL, OUTPUT); digitalWrite(BL, LOW);
pinMode(CL, OUTPUT); digitalWrite(CL, LOW);

// BLDC high-side pins
pinMode(AH, OUTPUT);
pinMode(BH, OUTPUT);
pinMode(CH, OUTPUT);

// Button
pinMode(MODE_BTN, INPUT_PULLUP);

// Buck PWM init
initBuckPwm();
stopBuckPwm();

allOff();
Serial.println("Starting...");
delay(500);
}

// =====
// LOOP
// =====
void loop() {
  checkModeButton();

  // Once buck mode is entered, BLDC commutation stops
  if (buckMode) {
    return;
  }

  unsigned long now = micros();

  if (now - lastStepTime >= (unsigned long)stepDelay) {
    lastStepTime = now;

    commutate(step);
    updateRamp();

    step++;
    if (step >= 6) step = 0;
  }
}

```

```

if (ramping && rampCount % 50 == 0) {
    Serial.print("delay=");
    Serial.print(stepDelay);
    Serial.print(" duty=");
    Serial.println(pwmDuty);
}
}
}

```

MATLAB code for simulations:

```

%% Regenerative Braking Simulation Using a Supercapacitor
% This simulation models a hub motor acting as a generator while slowing
% down a flywheel and charging a supercapacitor.

clear; clc; close all; % Clear workspace, command window, and close figures

% SYSTEM PARAMETERS

Iwheel = 0.006;    % Moment of inertia of wheel + motor (kg*m^2)

b = 0.001;        % Friction coefficient representing bearing and air drag

Ke = 0.255;       % Back EMF constant (V per rad/s)

Kt = 0.255;       % Torque constant (Nm per Amp)

R = 3;            % Motor winding resistance (Ohms)

C = 10;           % Supercapacitor value (Farads) - easy to change

% INITIAL CONDITIONS

RPM0 = 1500;      % Assumed starting speed of the wheel (RPM)

omega0 = 2*pi*RPM0/60; % Convert RPM to rad/s

```

```

Vcap0 = 0;          % Initial capacitor voltage (Volts)

x0 = [omega0; Vcap0]; % Initial state vector [angular speed; capacitor voltage]

% SIMULATION TIME

tspan = [0 20];     % Simulate for 20 seconds

% SOLVE DIFFERENTIAL EQUATIONS

[t,x] = ode45(@(t,x) regen_dynamics(t,x,Iwheel,b,Ke,Kt,R,C), tspan, x0);

omega = x(:,1);     % Extract angular speed from solution

Vcap = x(:,2);      % Extract capacitor voltage from solution

% CALCULATE REGENERATIVE CURRENT

Vemf = Ke .* omega; % Generated voltage from motor rotation

Iregen = (Vemf - Vcap) ./ R; % Current flowing into the capacitor

Iregen(Iregen < 0) = 0; % Prevent reverse current when voltage is too low

% PLOT 1: ANGULAR SPEED

figure

plot(t, omega, 'LineWidth', 2)

xlabel('Time (seconds)')

ylabel('Angular Speed (rad/s)')

```

```
title('Angular Speed vs Time During Regenerative Braking')
```

```
grid on
```

```
% PLOT 2: REGENERATIVE CURRENT
```

```
figure
```

```
plot(t, Iregen,'LineWidth',2)
```

```
xlabel('Time (seconds)')
```

```
ylabel('Current (Amps)')
```

```
title('Regenerative Current vs Time')
```

```
grid on
```

```
% PLOT 3: SUPERCAPACITOR VOLTAGE
```

```
figure
```

```
plot(t, Vcap,'LineWidth',2)
```

```
xlabel('Time (seconds)')
```

```
ylabel('Capacitor Voltage (Volts)')
```

```
title('Supercapacitor Voltage vs Time')
```

```
grid on
```



```
% DYNAMICS FUNCTION
```

```
function dx = regen_dynamics(~,x,Iwheel,b,Ke,Kt,R,C)
```

```
omega = x(1); % Current angular speed
```

```
Vcap = x(2); % Current capacitor voltage
```

```
Vemf = Ke * omega; % Voltage generated by motor rotation
```

```
I = (Vemf - Vcap) / R; % Current produced by generator
```

```
if I < 0
```

```
    I = 0; % Stop regeneration if generator voltage is too low
```

```
end
```

```
Tregen = Kt * I; % Regenerative braking torque
```

```
domega = (-Tregen - b*omega) / Iwheel; % Angular deceleration equation
```

```
dVcap = I / C; % Capacitor voltage change equation
```

```
dx = [domega; dVcap]; % Return derivatives to ODE solver
```

```
end
```